CASPA CLINTON

AI Engineer | AI Systems Architect | LLM Applications & Backend Engineering

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PROFESSIONAL SUMMARY

AI Engineer and Solutions Architect with 6+ years of experience building cloud-based applications, production backend systems, and AI-powered platforms. Specializes in Python, LLM application development, retrieval-augmented generation, agentic workflows, and scalable API architecture. Combines hands-on AI engineering with enterprise software architecture, cloud infrastructure, and technical leadership. Built AI applications featuring source attribution, evaluation workflows, and production-oriented guardrails.

TECHNICAL SKILLS

AI/ML & LLM Engineering: Python, PyTorch, Hugging Face, PEFT, LoRA, QLoRA, OpenAI API, Anthropic Claude API, RAG, hybrid search, semantic search, embeddings, prompt engineering, LLM evaluation, agentic architectures, multi-agent orchestration, Model Context Protocol (MCP)

AI Frameworks & Evaluation: LangGraph, LangChain, LlamaIndex, ChromaDB, Ragas, DeepEval, Langfuse, LangSmith, Streamlit, structured outputs, function calling, model evaluation, observability

Backend & Distributed Systems: FastAPI, Django, REST APIs, Python async programming, microservices, distributed systems, real-time applications, Pydantic, PostgreSQL, SQL, NoSQL, JavaScript, TypeScript

Cloud, DevOps & Infrastructure: Google Cloud, Microsoft Azure, Docker, Kubernetes, CI/CD, Vercel, cloud architecture, serverless deployment, infrastructure monitoring

Frontend & Application Development: React, Next.js, Flutter, Dart, TypeScript, Tailwind CSS

Engineering & Architecture: System design, enterprise architecture, technical leadership, software architecture, API design, developer mentoring, Git, automated testing

PROFESSIONAL EXPERIENCE

Caasitech Group , Yaoundé, Cameroon | Lead Software Engineer & Solutions Architect | 2021–Present

- Architected cloud-based microservices for SOS Cameroon, SOS Nigeria, and Kamerlink, delivering public safety and service marketplace applications across web and mobile platforms.
- Optimized backend performance by reducing API latency from 2.1 seconds to 0.4 seconds through scalable service architecture and performance improvements.
- Engineered cloud infrastructure across Google Cloud and Azure, achieving 99.9% platform uptime through resilient deployment and monitoring practices.
- Implemented CI/CD pipelines to reduce release times by 40% and improve deployment reliability across application environments.
- Integrated AI and predictive analytics into business applications to support data-driven features and improve user engagement.
- Developed an AI-powered employee experience platform that contributed to a reported 17% increase in company profitability through data-driven insights.
- Reduced infrastructure downtime by 20% through proactive monitoring, automation, and preventive maintenance systems.
- Built DevOps tooling that reduced incident response time by 60% through improved operational workflows.
- Mentored 8+ developers across web, mobile, and machine learning, supporting progression into more senior engineering responsibilities.
- Presented AI product innovation at Web Summit Qatar 2024, representing the company and communicating technical capabilities to an international audience.

Caasitech Academy | Technical Immersion Program | Feb 2019 – Oct 2020

- Developed software engineering foundations through hands-on projects in Python, Java, JavaScript, and TypeScript.
- Applied cloud architecture principles across Azure, microservices, and containerization to build practical software systems.
- Studied machine learning integration through applied technical projects and engineering exercises.

SELECTED AI ENGINEERING PROJECTS

The Persuasion Arena, AI Debate & Negotiation Coach

Stack: Python, OpenAI API, Streamlit

- Engineered an interactive AI debate coach that generates opposing arguments and evaluates user responses through structured conversational workflows.

- Integrated LLM-based scoring to provide feedback on persuasion and negotiation performance through an interactive application.
- Deployed a working AI application using Streamlit to make the system accessible through a browser-based interface.

Case File AI, Multi-Source Investigation Assistant
Stack: Python, OpenAI API, ChromaDB, Streamlit

- Built a retrieval-augmented investigation assistant that processes multiple sources and generates answers with source attribution.
- Implemented vector-based document retrieval using ChromaDB and embeddings to support question answering over unstructured information.
- Engineered source disagreement detection to flag conflicting information and improve the transparency of generated answers.

Apex, Autonomous Sales Agent
Stack: Python, LangGraph, OpenAI API, ChromaDB, Streamlit

- Designed a multi-stage autonomous sales workflow that sources, qualifies, and progresses leads through agentic orchestration.
- Implemented stateful agent execution using LangGraph to coordinate multiple stages of the sales process.
- Integrated human escalation logic to support human oversight when autonomous execution reaches defined decision points.

AI Engineering & Evaluation Portfolio, LLM Systems, RAG, Agents & Production Infrastructure
Stack: LangGraph, Ragas, DeepEval, Langfuse, LangSmith, MCP, Hugging Face PEFT

- Designed stateful agent workflows using LangGraph, including checkpointing and cyclic graph execution.
- Implemented retrieval pipelines combining semantic search, BM25, re-ranking, and query decomposition.
- Developed LLM evaluation workflows using Ragas and DeepEval to assess answer relevance and faithfulness.
- Integrated observability tooling with Langfuse and LangSmith for LLM tracing, latency logging, and token cost tracking.
- Built MCP tool servers exposing resources, prompts, and tools for integration with AI assistants and agentic workflows.
- Applied parameter-efficient fine-tuning using Hugging Face PEFT, LoRA, and QLoRA for domain-specific model development.

EDUCATION

Caasitech Academy | Hands-On Technical Immersion Program | Feb 2019 – Oct 2020
Focus: Cloud Architecture, Azure, Microservices, Containerization, Machine Learning, Python, Java, JavaScript, TypeScript.